Interpreting Agent Behavior (IAB):

Human-Centered Interpretation for Understanding Agents, Humans, and Interaction

NeurIPS 2026 Workshop Proposal · https://gaojie058.github.io/iab-workshop-proposal/

Tagline: IAB turns agent runtime data into human understanding of what agents do and how people work with them.

Background. Commercial autonomous agents such as Claude and Codex now run for hours or even days to complete tasks, and along the way they show complex behavior: they plan, reason, use tools, recover from errors, coordinate with subagents, and communicate with users [37, 42]. We use the word *behavior*, as in the study of human behavior [13, 36], for the full range of what an agent does during runtime. This behavior spans three levels: what agents do and how they do it (decomposing tasks, deciding under uncertainty), what people do in response (instructing, verifying, stepping in), and how the two work together through instructions and corrections. All three levels are generating vast behavioral data such as execution logs and interaction traces [28]. Yet existing approaches read this data largely for outcomes, not through a behavior lens [31]: benchmarks tell us whether an agent succeeds or fails [21–23, 29, 40, 45], but not *what* it did or *how* it did it. Understanding *what* and *how* is what people actually need. It lets agent developers and model trainers debug failures, compare architectures, and filter training data. It lets agent users and deployment engineers watch production agents to understand safety, cost, and reliability risks. Our purpose with IAB is to identify the emerging problem spaces and challenges, pushing the community towards building this missing layer between raw behavioral data and meaningful human *oversight*: debugging, governance, and calibrated trust all depend on first understanding what an agent did and how.

Understanding behavior at scale. The field lacks the vocabulary, methods, and tools to describe and analyze agent runtime behavior, even though the data is already here: the reasoning trajectories and traces produced while agents run. Early work has used grounded theory to interpret failure modes by hand [15], but failure is only one kind of behavior among many, alongside success, hesitation, and recovery. Understanding the full range means reading the running trajectories themselves, which are unstructured and large-scale. Humans cannot read through thousands of log entries; they need patterns, summaries, and explanations, in other words, *interpretation*. And even when we interpret by hand, we do not yet know how to scale it.

What IAB does. IAB therefore works *toward an interpretive science of agent behavior*. It treats behavior across these three levels as the object of study, and proceeds in two steps: first gathering the community to identify the problem space and emerging challenges within the domain. To analyze behavior, IAB uses the broad set of methods that social scientists have developed, such as grounded theory (refined over more than sixty years), to read meaning from data, discover categories from it, and then count them (taxonomies, frequencies, inter-rater reliability). IAB welcomes various methods that helps us understand behavior. In particular, IAB asks three questions:

> AGENTS **What do agents do, and how?** Which behaviors recur at runtime, and which are new: planning, reasoning, calling tools, handling ambiguity and uncertainty, failing, and recovering? What emergent behaviors appear in single- and multi-agent systems, and how does the choice of model or architecture shape them? What types does this behavior fall into, and are they similar enough to human behavior that we can borrow methods from studying it? How do we capture and reuse successful patterns?

> HUMANS **What do people do in response, and how?** What do people do when they work with agents: writing prompts, verifying outputs, over-relying, forming mental models of agent capabilities, and monitoring ongoing execution? What strategies do they use to adjust agent behavior mid-execution? What types does this behavior fall into, and is it similar to how they treat other people?

> INTERACTION **What happens when humans and agents interact, and how?** Which behaviors recur, which are new, and what role do people play? How do users communicate intent and goals, and how are misunderstandings and breakdowns repaired? What patterns emerge from interaction traces and logs, and how do they differ across tasks? Are these interactions similar to how people work with other people?

Why now? Real-world agent deployments have reached unprecedented scale by mid-2026, and their behavioral data is increasingly diverse: execution traces [19, 30], tool-use logs [27, 35], decision records [38, 41], and simulated behaviors [16, 33]. Researchers are actively developing empirical methods to make sense of this data, including grounded theory [15, 32], qualitative analysis [24, 26], error analysis [15, 17], corpus analysis [39, 43, 44], trace analysis [12, 14, 18, 20, 30], and red-teaming [34].

Building community. IAB bridges two communities: social science and HCI contribute the interpretation methods (e.g., computational social science, behavioral science, collective intelligence), and AI contributes the problem space (e.g., evaluation, governance, alignment, responsible AI). To make this community last, accepted papers form a non-archival **proceeding**, keynotes and the panel are recorded and made public, and we maintain a public **repository** of behavioral datasets, annotation schemas, and analysis tools, seeded by our organizers' *How to Interpret Agent Behavior* [25] runtime taxonomy and open analysis pipeline. We will also synthesize shared vocabulary, open problems, and methodological guidelines from submissions and the panel into a community-authored paper, a lasting foundation for the emerging field of agent behavior interpretation.

Expected attendance. We anticipate around 50 submissions and around 100-200 attendees. Closely related agent and interpretability workshops have drawn comparable submission volumes, and agent-focused workshops such as *AgentWild* (ICLR 2026) [8], *MTI-LLM* (NeurIPS 2025) [5], and earlier agent workshops (LLM Agents at ICLR 2024, Foundation Model Agents at NeurIPS 2024) have drawn 200+ attendees.

Submission and review process. We solicit non-archival **long papers** (up to 9 pages) and **short papers** (up to 4 pages), plus unlimited references, in NeurIPS style [11], across four contribution types: **1) Datasets and Resources** (annotated interaction traces, failure catalogs, human-agent conversational data); **2) Methods and Tools** (qualitative coding frameworks, think-aloud protocols, interaction visualization, LLM-as-judge pipelines); **3) Empirical Studies** (error taxonomies, failure-mode analysis, case studies of agent decisions, user studies); and **4) Human-Agent Interaction Analysis** (verification strategies, intent grounding, mid-execution adjustment, collaborative failure recovery). We also welcome negative results and position papers. Submissions are double-blind and receive three reviews each via OpenReview. Beyond the 8 organizers, who serve as primary reviewers and area chairs, we have confirmed 6 program chairs from our advisory board's networks to keep per-reviewer load manageable. We will also recruit around 15 more program committee members from our advisory board's students or postdocs to encourage a diverse reviewer pool.

Timeline: Submission deadline: August 29, 2026 (AoE) · Notification to all contributors (talks and posters): by September 29, 2026 (AoE) · Camera-ready: November 20, 2026 · Workshop: one of the official NeurIPS 2026 workshop days (December 11–13, 2026, depending on location).

Schedule: a one-day, in-person workshop; the full program is below (Tentative).

- **09:00–09:10** Opening Remarks
- **09:10–09:45** Keynote 1: Armando Solar-Lezama
- **09:45–10:20** Keynote 2: Diyi Yang
- **10:20–10:50** Contributed Talks (2 × 15 min)
- **10:50–11:40** Poster Session #1 + Coffee Break
- **11:40–12:10** Contributed Talks (2 × 15 min)
- **12:10–13:10** Lunch
- **13:10–13:45** Keynote 3: Bowen Baker
- **13:45–15:00** Contributed Talks (5 × 15 min)
- **15:00–15:50** Poster Session #2 + Coffee Break
- **15:50–16:35** Panel: *Empirical Methods for Understanding Agent Behavior*
- **16:35–16:50** Best Paper Award + Closing Remarks

Speakers & panelists (all confirmed, in-person): **Armando Solar-Lezama** (MIT CSAIL): Professor; program synthesis and neurosymbolic programming. **Diyi Yang** (Stanford): Assistant Professor; human-centered NLP and human-AI interaction. **Bowen Baker** (OpenAI): Research Scientist; multi-agent systems and emergent behavior. **Graham Neubig** (CMU LTI): Associate Professor; language agents and code intelligence; creator of OpenHands. The three keynotes (Solar-Lezama, Yang, Baker) also serve on the panel (*Empirical Methods for Understanding Agent Behavior*), joined by Neubig; additional panelists to be invited.

Supporting junior researchers and underrepresented groups. We accept non-archival submissions to lower the barrier for early-stage work, reserve spotlight slots and best-paper awards for student-led papers, encourage keynote speakers and panelists to share mentorship advice, and support remote attendance for those who cannot travel. Our funding efforts and broader inclusion measures are detailed in the Diversity, Inclusion, and Community Building section.

Location preference (most to least preferred): Sydney; Atlanta; Paris.

Logistics and special requirements. Venue-specific logistics will be finalized once the venue is assigned in the metareview; the permitted up-to-1-hour remote-presentation allowance serves as a contingency. We need standard AV (projector, screen, microphones), reliable Wi-Fi for live agent demonstrations, and roughly 20 poster boards; no nonstandard equipment is required.

Conflicts of interest. We follow the NeurIPS COI policy [10]: reviewing is double-blind through OpenReview with automatic conflict detection, no organizer or reviewer handles a submission from their own institution or a personal conflict, organizers and their students or postdocs may not submit, and organizers present no contributed work.

Diversity of organizers, speakers, advisory board, and committee. The organizing team of 8 members spans 5 institutions across academia and industry (Johns Hopkins University, Caltech, University of Hong Kong, Google DeepMind, xAI), with balanced gender representation (4 women, 3 men, 1 LGBTQ+ member), multiple career stages (assistant professors, postdocs, PhD student), and backgrounds from multiple countries, reflecting the global nature of the research community we aim to serve. Our invited speakers and panelists likewise span career stages and institutions (MIT, Stanford, OpenAI, CMU) and include women and early-career faculty. The 8-member advisory board carries this further, spanning academia and industry across five institutions (Johns Hopkins, Notre Dame, Northeastern, xAI, Bloomberg AI), career stages from assistant professors to a postdoc and industry engineers. The program committee draws reviewers from the NLP, HCI, and ML systems communities across institutions including UC San Diego and Kean University.

Funding and sponsorship: We are pursuing external support: we have begun discussions with advisory board member Fan Bai (Bloomberg AI) about sponsorship from Bloomberg, and are applying for a grant of approximately $5,000 for student travel grants and registration waivers for underrepresented participants.

Points of difference and previous workshops

IAB stands apart in both *problem scope* and *method*: it draws on social science and HCI to build the missing layer between raw behavioral data and human understanding. **At NeurIPS**, prior agent workshops mostly build and score agents rather than interpret them. *MTI-LLM* (NeurIPS 2025) [5] treats multi-turn interaction as a capability to improve, *Evaluating Agentic Systems* (NeurIPS 2025) [2] designs benchmarks and metrics for reliability and safety, *DL4C* (NeurIPS 2025) [4] builds and stress-tests coding agents, and *Open-World Agents* (NeurIPS 2024) [1] centered on agent construction and capabilities. These ask whether an agent succeeds and how to make it better; IAB asks what agents *actually do* at runtime and how to read it systematically, providing the methods that turn behavioral data into human understanding. **Elsewhere**, the closest is the *Workshop on Agent Behavior* (COLM 2026) [7], which studies how agents *do and should* behave but stays at the agent level; IAB adds human-centered interdisciplinary methods and extends to the human and interaction levels. The rest differ along the same line: *Actionable Interpretability* (COLM 2026) [6] and *BlackboxNLP* (EMNLP 2025) [3] look inside the model, *AgentWild* (ICLR 2026) [8] focuses on safety and security, and *HEAL* (CHI 2026) [9] judges model outputs.

Concurrent proposals. Three of our organizers with more experience organizing such events are concurrently proposing unrelated workshops (Teresa Yeo: CAO 2026; Daniel Khashabi: TTCL 2026; Kaiser Sun: STEPS 2026), but the lead organizer and the four remaining organizers are not proposing any other workshop at NeurIPS 2026, so their commitment to this one is clear.

This is the first edition of this workshop.

Organizer Information

Our 8-member committee from 5 institutions ensures diversity across multiple dimensions: **Career stage:** 2 assistant professors, 3 postdoctoral researchers, 1 PhD student, and 2 industry research scientists. **Sector:** both representation in academia (6) and industry (2). **Research background:** a mix of AI/NLP researchers and HCI researchers. **Geography:** members span institutions across the US, Singapore and Hong Kong. **Gender:** the committee includes 4 women, 3 men and 1 non-binary individual. Several core organizers have recently co-authored *How to Interpret Agent Behavior* [25], work that sits directly on the workshop's theme and shows the team's cohesion and shared expertise.

Jie (Sophia) Gao (`jgao77@jh.edu`) is a Malone Postdoctoral Fellow at Johns Hopkins University, affiliated with the Malone Center for Engineering in Healthcare, the Data Science and AI Institute, and the Center for Language and Speech Processing. Her research studies how humans perform interpretive work and how AI can support it, across qualitative analysis, code comprehension, and agent behavior analysis. Her recent work on CodeMap received an ACM SIGSOFT Distinguished Paper Award at ICPC 2026, and her earlier work on CoAIcoder (TOCHI 2023) and CollabCoder (CHI 2024) introduced widely adopted designs for human-AI collaboration in qualitative analysis. She co-organized the CHI 2024 workshop "LLMs as Research Tools" and built mindcoder.ai, a public platform for AI-assisted qualitative analysis. *Contribution: Jie brings expertise in AI-assisted qualitative analysis and HCI research methods to agent behavior analysis.*

Zhuoran Lu (`zhuoranlu33@gmail.com`) is an incoming Assistant Professor at the University of Hong Kong, conducts research on human-AI interaction and human-centered AI, with a focus on how people interpret, trust, and rely on AI-based systems. His work has been published in leading HCI venues, including CHI, CSCW, and IUI, as well as AI venues such as ICLR, AAAI, and EMNLP, and has received recognition, including a CSCW 2022 Best Paper Award. *Contribution: Human-AI interaction and behavioral science expertise; human-centered evaluation design.*

Katherine Van Koevering (`kvankoe1@jh.edu`) is a postdoctoral fellow at the Johns Hopkins Data Science and AI Institute. She received her PhD from Cornell University. Her research focuses on the flow of people and ideas through networks and hidden linguistic gradients in digital communication. *Contribution: Katherine brings expertise in computational social science, computational linguistics, and digital communication analysis to the workshop, providing insights into how information propagates through human-agent interactions.*

Sijie Ji (`sijieji@caltech.edu`) is a Schmidt Science Fellow at Caltech. She develops engineering principles that expand the observability of real-world systems, allowing AI to reason about human behavior, physiology, and hidden internal states beyond traditional sensing methods. Dr. Ji has served as an organizer and program committee member for conferences such as SenSys, BuildSys, MobiSys, MobiCom, and CPS-IoT Week. She has been recognized as an NSF CPS Rising Star, an NIH mHealth Training Institute Scholar, and a recipient of the N2Women Young Researcher Fellowship. *Contribution: Sijie brings expertise in cyber-physical systems, real-world observability, and AIoT deployment, contributing a systems perspective on how agent behavior should be interpreted under real-world constraints, noisy interaction environments, and human-in-the-loop workflows.*

Boyuan Zheng (`steven.zheng010@gmail.com`) is a Member of Technical Staff at xAI. His research focuses on building language agents for digital environments, including scaling RL training for computer-use agents, self-play, and automated research. He's a core contributor to the computer-use and web dev capabilities of Grok 4.20 and has published widely on computer-use agents, including MolmoWeb, SeeAct, UGround, and Mind2Web. He co-organized the Workshop on Computer Use Agents at ICML 2025. *Contribution: Boyuan brings expertise in building and training language agents for digital environments, providing insights into real-world agent behavior from an industry deployment perspective.*

Teresa Yeo (`teresa.ysq@gmail.com`) is a Research Scientist at Google DeepMind. Her research focuses on understanding and steering foundation model behavior under distributional and task shifts, with particular interest in how models can be monitored and adapted when their behavior departs from expectations, a critical challenge as models become agentic. She co-organized the 2nd and 3rd Test-time Adaptation Workshop (ICML'25 and ICLR'26), the 1st Workshop on Monitoring ML Models Under Drift (ICLR'26), and was an invited speaker at the 1st Test-time Adaptation Workshop (CVPR'24). *Contribution: Teresa brings expertise in model behavior monitoring and adaptation to the workshop, with a focus on real-world deployment challenges. Concurrent proposal: the 2nd Workshop on Catch, Adapt, and Operate (CAO).*

Daniel Khashabi (`danielk@jhu.edu`) is an assistant professor in the Department of Computer Science at Johns Hopkins University and is affiliated with the Data Science and AI Institute and the Center for Language and Speech Processing (CLSP). His research is anchored around natural language as a key medium for communication, driven by the goal of making these systems more helpful, reliable, and efficient. He is the recipient of AI2's Lasting Impact Award (2024), multiple industry faculty awards (Amazon, Google, Apple), and several Best Paper Awards (COLM 2024, ACL 2023, NAACL 2022). His teaching has awarded him the Joel Dean Award for Excellence in Teaching in the Department of Computer Science in 2025. He has experience organizing Workshop on Instruction Tuning and Instruction Following (NeurIPS 2023) and Student Research Workshop (ACL 2019). *Contribution: Daniel brings expertise in language-driven reasoning systems and their evaluation to the workshop, with a focus on reliability and real-world impact. Concurrent proposal: "TTCL 2026: Towards Test-Time Continual Learning Agents".*

Kaiser Sun (`hsun74@jhu.edu`) is a PhD student at the Center for Language and Speech Processing and Data Science and AI Institute at Johns Hopkins University, advised by Mark Dredze. Their research focuses on the training dynamics, interpretability, and multimodality of LLMs. Their work has been recognized with a Best Paper Honorable Mention at CoNLL 2023 and coverage by MIT Technology Review. Previously, they gained industry experience at Meta AI and AWS AI Labs. *Contribution: Kaiser brings expertise in LLM interpretability and training dynamics to agent behavior analysis. Concurrent proposal: Workshop on the Science of Training: Emergence, Prediction, and Steering (STEPS)*

Program Committee

We have so far confirmed 8 PC members from the NLP, HCI, and ML systems communities: Heyuan Huang (Johns Hopkins University), Arman Hatami (Johns Hopkins University), Yadi Cao (UC San Diego), Boyang Li (Kean University), Alyssa Columbus (Johns Hopkins University), Han Jiang (Johns Hopkins University), Yifan Zhang (National University of Singapore), and Huiqi Zou (Northeastern University). Together with our advisory board, we will recruit more to reach 20-30 reviewers in total, using our own and the advisory board's networks. This keeps each submission at 3 expert reviews and no reviewer at more than 3 papers.

Advisory Board

Throughout planning and running the workshop, our advisory board gives us visibility in their communities, connects us to potential funding sources, speakers, and attendees, and helps us recruit more program committee members.

- **Ziang Xiao**, Assistant Professor, Johns Hopkins University: human-AI interaction
- **Soufiane Hayou**, Assistant Professor, Johns Hopkins University: deep learning theory
- **Toby Jia-Jun Li**, Assistant Professor, University of Notre Dame: interactive AI systems
- **Hang Jiang**, Assistant Professor, Northeastern University: NLP, social AI
- **Weiyan Shi**, Assistant Professor, Northeastern University: AI agents, human-AI coevolution
- **Samuel Nathanson**, xAI: industry perspective on agent deployment
- **Fan Bai**, Senior Software Engineer, Bloomberg AI: agent safety and optimization; industry perspective and sponsorship liaison
- **Jen-tse Huang**, Postdoctoral Researcher, Johns Hopkins University: LLM behavioral evaluation expertise.

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[4] DL4C: Deep learning for code workshop at NeurIPS 2025. https://dl4c.github.io/, 2025.

[5] MTI-LLM: Multi-turn interactions of LLMs workshop at NeurIPS 2025. https://workshop-multi-turn-interaction.github.io/, 2025.

[6] Workshop on actionable interpretability at COLM 2026. https://actionable-interpretability.github.io/, 2026.

[7] Workshop on agent behavior at COLM 2026. https://www.aiagentbehavior.com/, 2026.

[8] AgentWild: Workshop on agent safety and security at ICLR 2026. https://agentwild-workshop.github.io/, 2026.

[9] HEAL: Human-centered evaluation of language models workshop at CHI 2026. https://heal-workshop.github.io/, 2026.

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